

Embeddings and neighbourhoods

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Sequence representations

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Training a small image model

Without opening the old lesson, explain one key idea from Training a small image model. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image tensors and channels

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Calculate distances and a tie for Embeddings and neighbourhoods. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Embeddings and neighbourhoods. Proposed explanation: “Token ID differences define embedding distances”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Find a neighbourhood independently

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query. Can that mean identify which ordered pair produced it?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace the lookup shape

Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

My reasoning and evidence

6. W2 Calculate distances and a tie

For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

My reasoning and evidence

7. W3 Trace a trainable row

For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

My reasoning and evidence

8. W4 Find a neighbourhood independently

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query. Can that mean identify which ordered pair produced it?

My reasoning and evidence

9. W5 Keep the lookup consistent

A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled embeddings and neighbourhoods evidence sheet to Image-pattern Network Investigation. Show how you can look up a vector using a token id and preserve sequence shape. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

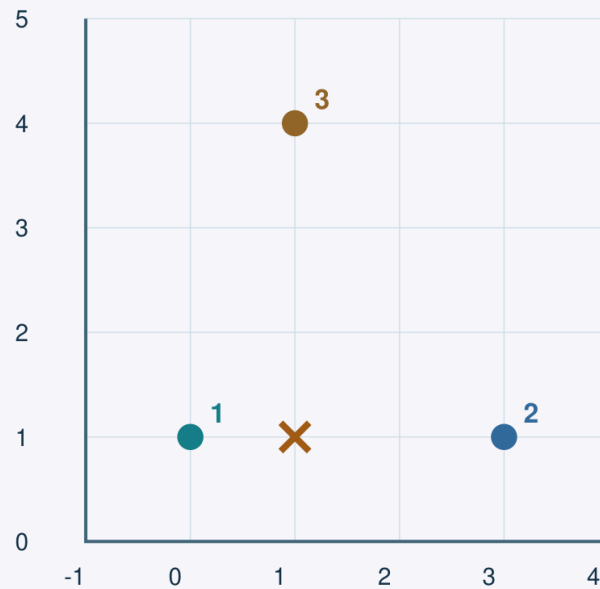
12. Pictorial case practice: Embeddings and neighbourhoods

Measure embedding neighbours without turning numerical proximity into factual certainty.

Toy table $P=[0, 1]$, $Q=[3, 1]$, $R=[1, 4]$; query $[1, 1]$. Use squared Euclidean distance.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

Embeddings and neighbourhoods

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Distance to P

$$(1-0)^2 + (1-1)^2 = 1.$$



2. Distances to Q/R

Q distance 4; R distance 9.



3. Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

CHANGE ONE THING

Move the query to [2, 1].

Explain the changed result using the matching lesson.

What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Prepare the demonstration. Gather reproducible instructions, evidence, access checks and remaining limitations.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Measure embedding neighbours without turning numerical proximity into factual certainty.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? P is nearest under the declared metric; this is a geometric result, not proof of meaning.

1. Distance to P
2. Select neighbour
3. Distances to Q/R

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Select neighbour P is nearest under the declared metric; this is a geometric result, not proof of meaning. An embedding stores each item as an ordered vector. Under squared Euclidean distance, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector. These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs evidence about how the representation was learned and used.

Practice 2: explain the deciding evidence

Changed-case practice: Move the query to [2, 1]. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. Q becomes nearest with squared distance1; P distance4; R distance10.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Q becomes nearest with squared distance1; P distance4; R distance10. Changing the query position changes geometric neighbours while the stored table remains fixed.

Repair a missing building block

Review Sequence representations (ai-5/25). Goal: Encode an ordered sequence with an explicit vocabulary. Short practice: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Training a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower training loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
